

ERASTEEL TECHNICAL DATASHEET: ASP 2012

Application Family: Cold work | Standards: Erasteel Cold Work PM Steel

1. Grade Overview & Characteristics

ASP 2012 is a premier cold work powder metallurgy steel engineered for exceptional impact toughness, fatigue resistance, and chip resistance under severe shock loading. Ideal where standard tool steels suffer catastrophic breakage.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	0.58 - 0.62	3.8 - 4.2	1.9 - 2.1	2.0 - 2.2	1.4 - 1.6	0.0	Ultra-clean high ductility microstructure

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	58.0 - 61.0 HRC	Machinability	7.5/10
Max Reachable Hardness	61.0 HRC	Wear Resistance	5.5/10
Annealed Delivery State	Max 280-350 HB	Toughness	9.5/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	5.0/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	8.0/10

4. Industrial Applications

Heavy blanking punches, high-impact cold forming dies, cold extrusion tools, shear blades for thick plate, plastic injection molds.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1050-1120°C. Double/triple tempering at 540-570°C. Working hardness 58 to 61 HRC.